Can AI Swing? A Musician-Centred Evaluation of Jazz Competence in Audio Generation Models

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Abstract: AI music generation systems can produce audio that superficially resembles jazz. It remains unclear whether these models achieve genuine musical competence or only reproduce surface-level sound. Jazz is an ideal genre of music to test the difference as its rules are deep and codifiable: swing microtiming, functional harmony, and formal structure, while remaining learnable from a large recorded dataset, so that both success and failure are interpretable. We present a musician evaluation of AI-generated jazz spanning current-generation systems and core jazz styles: fourteen pieces are rated in depth against a six-axis rubric - harmonic authenticity, swing feel, improvisational coherence, idiomatic vocabulary, ensemble interaction, and formal structure - by a trained jazz pianist, and complemented by an automated analysis pipeline for onset-based swing-ratio detection and chroma-based chord detection. A consistent pattern is seen: the models show stronger local coherence than global form, with formal structure the weakest axis; automated swing-ratio detection bears no reliable relationship to perceived swing and misleads most at the extremes of its range; and template-based ii–V–I detection fails on mixed audio - findings that bear on evaluation methodology as much as on the models. The strongest piece was judged indistinguishable from amateur human jazz by a non-specialist listener, though not by a trained one. We contribute a replicable evaluation framework, a diagnostic axis-by-axis breakdown of jazz competence in current systems, and a critical analysis of the gap between automated audio metrics and musician-perceived quality.

Keywords: *AI music generation; Jazz analysis; music generation evaluation; swing ratio; expert evaluation; music information retrieval*

1. Introduction: Since the last three years, the scope for AI music generation has gone from a research curiosity to a consumer technology used by millions. Models such as Suno and Udio generate fully produced, multi-instrument pieces from a simple text prompt and other models span various architectures; right from language-model token prediction to latent diffusion to symbolic composition. Their outputs are increasingly difficult to distinguish from human

recordings on first listen, and they are marketed and consumed on the assumption that the music is not just convincing but actually competent.

That assumption hides an unresolved question. Sounding like a genre is not the same as being competent within it. A system can mimic a style's timbre, instrumentation, and surface gestures while failing at the structural logic in the music a trained musician treats as essential. The distinction separates two claims: that a model has learned the "sound" of a genre, and that it has learned the genre itself. Lay-listener studies and aggregate audio-similarity metrics cannot tell these apart, because both texture and underlying competence register as "plausible" to an untrained ear and a distributional metric alike.

Jazz is perfect for this distinction. It is deep but codifiable: swing as microtiming, functional harmony with ii-V-I motion and voicings, improvisation that develops in a proper manner, and clear structure such as the 32-bar AABA and the 12-bar blues. Yet it is also learnable in principle from the dataset these models train on. This makes both success and failure interpretable: a success identifies where competency was achieved, and a failure points to a specific structural gap rather than generic "wrongness." Jazz also has a documented perceptual asymmetry: trained musicians are able to hear flaws in generated jazz that normal people miss, so the evaluator is essential to the final result.

Therefore, we adopt a musician-centred evaluation. Instead of asking whether generated jazz can appeal to a general audience, we ask whether it withstands analysis from trained jazz musicians applying the same criteria used to assess human performance. This leads to a deep diagnostic analysis: a structured, axis-by-axis account of where competence is present and where it breaks down, paired with quantitative computer audio analysis.

The work addresses three questions:

1. Do current AI music systems produce jazz that is competent by music-theoretic criteria, or only superficially plausible?
2. Along which dimensions do they succeed and fail, and do the failures form a consistent pattern across pieces and styles?
3. How well do standard automated metrics for swing and harmony agree with expert judgment, and what does any divergence imply for how AI music should be evaluated?

The paper makes three contributions. First, a replicable evaluation framework: a fixed prompt set with six jazz styles, applied across current systems and scored on a six-axis rubric with proper criteria. Second, a diagnostic breakdown of jazz competence, with one consistent finding: these models show markedly stronger local coherence, in harmony and texture, than global form, which is the most reliable weakness across the pieces evaluated. Third, a critique of the evaluation instruments themselves: onset-based swing-ratio detection bears no reliable relationship to perceived swing in polyphonic audio and misleads most at the extremes, and template-based ii–V–I detection fails on mixed recordings: so the automated metrics diverge from musician-perceived quality in ways that are themselves informative. This is a caution for the growing practice of scoring AI music with off-the-shelf audio metrics, which are better treated as diagnostics than as ground truth.

2. Related Work

2.1 AI Music Generation Systems

There are three paradigms for contemporary music. The first is an autoregressive language model, using discrete audio tokens, used by MusicGen (Copet et al., 2023), which generates audio by predicting sequences of neural codec tokens and supports conditioning on both text and melody. The second is latent diffusion, in which audio is generated by iterative denoising in a learned latent space, used by Stable Audio and its successors (Evans et al., 2024) and they are notable for generating longer and timing-conditioned outputs. The third is symbolic generation, in which the model produces a score or MIDI representation rather than raw audio. The commercial systems evaluated here, Suno and Udio, are closed but are generally understood to combine language-model and diffusion techniques to produce fully mixed, multi-instrument songs from text prompts. The systems in our study are chosen to span these paradigms, since the way a model represents music internally is related to in the way that it fails.

Evaluation methods have not kept pace with capability. There are three main approaches to this:

1. Subjective listening tests that ask raters, usually non-specialists, to judge quality or preference. This can capture plausibility but rarely is it able to evaluate genre-specific competence, and their raters lack the training to hear the structural faults of the pieces.

2. Distributional metrics such as Fréchet Audio Distance (Kilgour et al., 2019) compare generated audio to a reference set which is useful for tracking quality, however, they are opaque as to which musical property differs and can reward surface realism over coherence.
3. Genre-classification accuracy measures surface signifiers, not idiomatic competence. For example, Prior work argues that such distributional measures are necessary but insufficient (Yang & Lerch, 2020), and jazz-specific analysis using pitch-class, groove, and chord statistics shows a research-grade generative model falling short of human jazz on exactly these dimensions (Wu & Yang, 2020)

We extend this line from older symbolic models to current commercial audio generators, and from computational features to expert perceptual judgment. None of these instruments distinguishes music that sounds like a genre from music competent within it, which is the exact distinction a trained musician can make.

2.2 Computational Analysis of Jazz

Our pipeline builds on prior quantification of jazz. On rhythm, Friberg and Sundström (2002) measured the swing ratio from professional drummers' ride patterns, finding it varies with tempo, from roughly 3.5:1 at slow tempo toward 1:1 at fast tempo, with the often-cited 2:1 "triplet" feel appearing only at a particular tempo. Benadon (2006) recast jazz eighth notes as an expressive microrhythm, showing swing is a flexible device musicians vary continuously rather than a fixed ratio. Both points motivate our attention to variance and to the gap between measured ratio and perceived swing.

On harmony, chroma-based analysis has dominated pitch extraction since Fujishima (1999) introduced pitch-class profiles for chord recognition. Later work improved estimation of the difficult, extended chords that characterise jazz, such as Mauch & Dixon (2010) using approximate note transcription, with hidden-Markov, structural, and deep-learning approaches following. Our pipeline uses chroma-based template matching in this tradition, with pattern

matching over the estimated chord sequence to detect chord progressions, the patterns most characteristic of functional jazz.

Two limitations of this literature are central to our contribution. Most swing and chord methods were developed on clean signals with solo instruments, isolated ride cymbals, and well-separated sources. Whereas AI-generated jazz is dense, fully mixed polyphony, in which harmonic smearing and onset ambiguity degrade both swing and chord detection. Section 4 shows these degradations are not incidental: the automated measures diverge from expert perception in systematic, interpretable ways, itself a result of applying classical methods to generate audio.

2.3 Human Evaluation in Music

Structured assessment of musical performance is well established outside AI. Graded frameworks such as ABRSM (Associated Board of the Royal Schools of Music, 2023) and Rockschool (RSL Awards, 2024) score performances against anchored, level-by-level criteria across dimensions, turning expert judgment into reproducible scores rather than global impressions; our six-axis rubric adopts a similar design.

Finally, training changes what a listener hears. Musicians can hear structural and idiomatic features that casual listeners miss, and prior work on AI jazz shows experts distinguishing machine from human material far more reliably than non-experts: in a Turing-style study, jazz experts identified computer-generated solos with 64.4% accuracy while non-experts managed 41.7%, below chance (Frieler & Zaddach, 2022). This difference is the core justification for our approach: if AI-jazz failures are audible chiefly to trained listeners, non-trained listener tests systematically overstate competence, and expert rubric-based evaluation is needed to characterise what these systems have learned. To our knowledge, no prior study applies expert-musician, rubric-based evaluation paired with quantitative audio analysis to the current generation of commercial audio jazz generators. That is the gap that is addressed here.

3. Methodology

The design pairs a structured expert evaluation with an automated pipeline, applied identically to every piece. The guiding principle is reproducibility: prompt design, generation,

rating, and analysis are each fixed in advance and documented, and all prompts, code, audio, and per-piece assessments are released in the accompanying repository. Given the scope of a small set of pieces assessed in depth by a single rater we frame the findings as diagnostic and pattern-level rather than as a powered comparison between systems.

3.1 Materials

We evaluate pieces spanning current systems and core jazz styles; fourteen are analysed by the pipeline and all fourteen are rated in depth. The four systems used are two prompt-driven commercial audio generators (Suno 4.5, Udio), a latent-diffusion generator (Stable Audio 2), and an open-source autoregressive language model over audio tokens (MusicGen, run as musicgen-large). The paradigms are chosen so the sample spans the dominant families of current audio generation — autoregressive token prediction, latent diffusion, and the hybrid commercial systems — rather than any one exhaustively.

3.2 Prompt Analysis

Each piece is generated from a fixed prompt targeting one style each: up-tempo bebop, modal jazz, jazz ballad, twelve-bar blues, big-band swing, and Latin jazz. These are chosen to exercise different competencies: bebop stresses harmonic and melodic vocabulary; modal jazz, restraint and development over static harmony; blues and big-band, form and swing; Latin jazz, rhythmic idiom. Each prompt specifies style, approximate tempo, and instrumentation but is deliberately under-specified, so the model's own choices remain visible. Full prompt texts appear in Appendix A. These prompts are always fixed and the exact same wording is used for all generations.

3.3 Evaluation Rubric

Each piece is scored on six axes, each an integer from 1 to 5 against anchored descriptors; half-scores are not used, and a judgment falling between levels takes the lower. The six scores sum to a total out of thirty. The full anchored rubric, with concrete anchors at each level, is in Appendix B. The six axes are Harmonic Authenticity, Swing Feel and Microtiming, Improvisational Jazz Vocabulary, Ensemble Interaction and Formal Structure.

Each score carries a written justification citing timestamps, so every rating traces to an observable event rather than a global impression. The axes and anchors are adapted from graded music-education frameworks and from the dimensions prior AI-music evaluation identifies as most diagnostic.

3.4 Generation Protocol

Generation follows a fixed protocol to prevent selection bias. The exact Appendix A prompt is used unmodified, and only each system's first output is retained — no regeneration, no cherry-picking. For every piece we record model name and version, generation date, exact prompt, and output duration, since commercial systems update without notice. Raw audio is saved before rating. When the systems provided 2 or more different outputs per prompt, a random one was picked.

3.5 Rating Procedure

Ratings come from a single expert rater: a jazz pianist with ten years' training (Grade 8 Rockschool) scoring against the anchored rubric, in a single sitting where possible to hold listening conditions constant. The rater is an author but was blind to model identity while scoring: a delay separated generation from rating and pieces were rated in random order. Every axis score is recorded with a timestamp-anchored justification, checkable against the released audio. We do not claim multi-expert consensus; the rater's expertise is the deliberate instrument of the study, on the premise — supported by Section 2.3 — that a trained musician detects competency failures lay listeners and metrics miss.

3.6 Quantitative Analysis Pipeline

An automated pipeline computes a fixed set of measures per piece. Swing ratio is estimated by detecting onsets with librosa (McFee et al., 2015), pairing consecutive onsets within an eighth-note window at the detected tempo, and taking the long-to-short ratio for each pair; we report mean, standard deviation, and a time-series plot, reading ~1.0 as straight, 1.5 as medium, 2.0 as hard triplet feel. Because onset detection is noisy on dense polyphony and most reliable for exposed melodic lines, the ratio is treated as indicative rather than definitive — a caution the results bear out. Chord vocabulary and harmony reduce a constant-Q chromagram to chord

estimates, classified as triad, dominant/major/minor seventh, half-diminished, or diminished seventh, with a residual for uncertain segments; candidate ii–V–I progressions are found by sliding a root-interval pattern over the sequence. Key is estimated via Krumhansl–Kessler profiles (Krumhansl & Kessler, 1982) on the mean chroma, and the pitch-class distribution is compared by cosine similarity to jazz and pop references. Self-similarity is computed from chroma to gauge structural repetition; it is a per-piece output, with systematic analysis deferred to future work. The pipeline outputs, per piece, a swing-ratio time series, a chord timeline, a self-similarity matrix, and a batch summary, enabling within- and cross-piece comparison.

3.7 Limitations

There are several limitations to this analysis.

1. The study rests on a single rater; every score is justified, but inter-rater reliability cannot be established, so the ratings are one expert's structured judgment rather than a consensus.
2. The sample is small: the findings are diagnostic and hypothesis-generating rather than statistically conclusive, especially for any comparison between systems.
3. Polyphonic audio degrades both swing and chord detection; this is documented in the results not as a caveat but as a finding, since the measures diverge from perception systematically.
4. The commercial models are black boxes that may change between generation and publication, so results describe the systems as they were on the generation dates.
5. A another limitation is methodological rather than statistical. The swing-feel rubric was revised once during the study: an earlier version of the anchor awarded a top score only when the measured swing ratio fell in the 1.5–2.0 range, and this criterion was removed once the analysis in Sections 4.4 and 5.3 showed the ratio to be an unreliable proxy for perceived swing. We flag this explicitly as a post-hoc change, motivated by a finding of the study rather than fixed in advance. Because the axis was otherwise scored purely by ear, the measured ratio entered scoring only through this single criterion; all swing-feel scores reported here were assigned, and subsequently re-confirmed, under the final,

ratio-independent rubric, so no reported rating depends on the removed criterion. Readers should nonetheless weigh the revision as a departure from a fully pre-registered protocol.

6. Finally, The two MusicGen pieces were generated with musicgen-large and are shorter than the commercial and diffusion outputs, and formal structure cannot fail over a span too short to contain a second section. The MusicGen formal-structure scores (both 1/5) should be read with this length difference in mind, though the rater's finding of near-total incoherence within the first phrase indicates the failure is not one of duration alone.

4. Results

The pipeline was run on fourteen pieces, and all fourteen are rated. They bring the study to four systems spanning the dominant generation paradigms — the two commercial hybrids (Suno 4.5, Udio), latent diffusion (Stable Audio 2), and open-source token prediction (MusicGen). Given the sample size, results are reported per piece and read as diagnostic patterns rather than powered comparisons; recurring patterns are noted as such. All per-piece reports and figures are in the accompanying repository.

4.1 Quantitative Summary

Table 1 reports the pipeline output for the fourteen analysed pieces, ordered by measured swing ratio, together with the number of valid onset pairs underlying each swing measurement: a reliability indicator that Section 4.4 shows to matter. Two features are immediately apparent. First, harmonic surface content is uniformly rich and jazz-aligned: jazz pitch-class similarity never falls below 0.91, the proportion of seventh-or-richer chords ("jazz complexity") ranges from 63% to 98%, and eight of the fourteen sit at 90% or above. Second, and in sharp contrast, template-based ii–V–I detection returns a total of two progressions across all fourteen pieces, one each for two pieces and zero in the other twelve. This is a striking result for music this harmonically dense, and is the first sign that the harmonic detector is failing rather than the music lacking functional motion (Section 4.4).

Table-1: Automated pipeline output (fourteen analysed pieces, ordered by swing ratio).

Piece	**Tempo**	**Key**	**Swing ratio (±sd)**	**Onset pairs**	**Jazz complexity**	**ii–V–I**	**Unique chords**	**Jazz PC sim.**	**Harm. complexity**	**Model**
Latin Bop	140	C♯ min	1.047 ± 0.268	785	88%	0	33	0.913	0.915	Stable Audio 2
musicgen_blue	121	C maj	1.071 ± 0.371	515	97%	0	40	0.983	0.959	MusicGen
Corner Pocket Riff	100	G maj	1.096 ± 0.422	404	98%	0	28	0.969	0.966	Suno 4.5
Sprint of the Brass	110	E ♭ maj	1.105 ± 0.449	562	90%	0	46	0.941	0.902	Stable Audio 2
musicgen_green	114	B ♭ min	1.108 ± 0.397	325	95%	0	39	0.941	0.937	MusicGen
Rail Yard Bop	125	C maj	1.159 ± 0.500	681	94%	0	53	0.984	0.932	Suno 4.5

Whispers of the Dorian Sky	134	C♯ min	1.225 ± 0.617	98	77%	0	40	0.948	0.859	Udio
Chasing the High Flight	141	F min	1.302 ± 0.687	482	94%	1	49	0.964	0.906	Udio
Blue and Purple	120	E ♭ maj	1.358 ± 0.747	115	72%	0	53	0.934	0.841	Stable Audio 2
Lush Chord Dreams	108	E ♭ maj	1.487 ± 0.859	192	76%	0	40	0.955	0.877	Udio
Moonlit Voicings	69	F maj	1.513 ± 0.819	161	70%	1	40	0.965	0.850	Suno 4.5
Blue Dorian Drift	128	E ♭ maj	1.710 ± 0.894	186	63%	0	58	0.941	0.836	Suno 4.5
The Flaming Band	152	B ♭ maj	1.786 ± 0.950	36	94%	0	55	0.961	0.917	Stable Audio 2

Blue Note Bounce	144	G min	2.014 ± 0.870	238	97%	0	47	0.965	0.947	Stable Audio 2

4.2 Rubric Scores

Table 2 gives the expert scores for the fourteen rated pieces. Totals span a wide range, from 10 to 26 out of 30 (mean 18.4). Across the fourteen, the weakest axis is formal structure (mean 2.3), consistent with the literature identifying large-scale form as the most reliable failure of generated music; the strongest is ensemble interaction (3.7), followed by swing feel (3.3), with harmonic authenticity, improvisational coherence, and idiomatic vocabulary clustered just below (3.0–3.1). Adding the two MusicGen pieces — both of which score at the floor of the range — pulls the overall mean down from 19.5 to 18.4 and sharpens the harmony axis's fall: harmonic authenticity, tied for strongest in the preliminary four-piece sample (3.5), now sits at 3.0, driven by pieces whose chord types are rich but whose chord motion is not. This dissociation is developed in Section 4.3.

Table-2: Analyzed rubric scores (fourteen pieces).

Piece	**Harm. auth.**	**Swing feel**	**Improv. coh.**	**Idiom. vocab.**	**Ensemble**	**Formal**	**Total**	**Model**
Rail Yard Bop	4	5	4	4	5	4	**26**	Suno 4.5
Corner Pocket Riff	3	5	3	4	5	5	**25**	Suno 4.5

Moonlit Voicings	4	4	4	4	4	3	**23**	Suno 4.5
Lush Chord Dreams	4	4	4	4	5	2	**23**	Udio
Blue Note Bounce	4	4	4	3	3	3	**21**	Stable Audio 2
Blue and Purple	4	3	3	3	5	2	**20**	Stable Audio 2
Sprint of the Brass	2	3	4	5	4	2	**20**	Stable Audio 2
Blue Dorian Drift	4	3	4	3	3	2	**19**	Suno 4.5
The Flaming Band	3	2	3	2	4	3	**17**	Stable Audio 2
Latin Bop	3	3	2	3	4	2	**17**	Stable Audio 2

musicgen_green	2	3	2	2	3	1	**13**	MusicGen
Chasing the High Flight	2	3	2	3	2	1	**13**	Udio
musicgen_blue	1	2	2	1	3	1	**10**	MusicGen
Whispers of the Dorian Sky	2	2	1	2	2	1	**10**	Udio
Mean (n=14)	3.0	3.3	3.0	3.1	3.7	2.3	*18.4*	

With model attributions confirmed for all fourteen rated pieces, a system-level pattern is visible, though with two to five rated pieces per system it can only be read as suggestive. The four Suno 4.5 pieces occupy the upper portion of the range (19–26), including the three highest totals in the study. The five rated Stable Audio 2 pieces cluster in the middle (17–21). The three Udio pieces span nearly the full range of the study (10–23): the two lowest Udio totals (*Whispers of the Dorian Sky*, 10; *Chasing the High Flight*, 13) sit at the floor, yet *Lush Chord Dreams* reaches 23, tied for third overall and carrying the strongest detectability verdict in the dataset — high variance within that system rather than a uniform level. The two MusicGen pieces are the lowest-scoring system in aggregate (10 and 13), consistent with an open-source token-prediction model trailing the tuned commercial systems, though two pieces cannot establish this. These are diagnostic observations on a small, unbalanced sample generated from different prompts per

system, not a controlled comparison; the fully model-crossed design identified in Section 6 is what would be required to make full system-level claims.

4.3 Axis-by-Axis Analysis

Harmonic authenticity: The expanded sample makes this axis the study's clearest dissociation between surface richness and functional logic. Six rated pieces score 4/5, and on these the rater consistently heard functional motion the detector missed: the full E ♭ maj7→G7→F♯maj7→Fm7→G♯m7→C♯7→Gm7→C7→Fm7→B ♭ 7 progression in Blue Dorian Drift (with Gm7→C7→Fm7 and Fm7→B ♭ 7→E ♭ as canonical ii–V–Is, detector count zero); ii–V–I as a defining feature of Rail Yard Bop; a near-complete C-minor palette (Cm7, E ♭ maj7, A ♭ maj7, G7, Dm7 ♭ 5) in Lush Chord Dreams; and a Bb7→G7→Cm7→F7 core in Blue Note Bounce, where the rater hears Cm7→F7 as a functional ii–V in B ♭ that the detector again scores as zero. The reverse pattern is equally diagnostic: Corner Pocket Riff posts a very high jazz-complexity figure (98%) yet scores 3/5, its harmony "generic … no extensions or tension"; Sprint of the Brass (90%) scores 2/5 with "no visible proper progression at all"; and Chasing the High Flight (94%) scores 2/5, with "essentially no chord progression" though the individual notes are correct. The MusicGen piece musicgen_blue is the extreme: it posts 97% jazz complexity and 0.983 pitch-class similarity — among the highest figures in the dataset — yet scores 1/5, the rater finding "completely random chords with nothing flowing into each other." Richness and function are fully dissociable here. A percentage cannot tell a piece that resolves from one that just plays chords.

Swing feel. Analysed in Section 4.4; in brief, the two pieces scored 5/5 — Rail Yard Bop and Corner Pocket Riff — carry two of the lowest measured ratios, while the three highest ratios in the dataset, Blue Dorian Drift, The Flaming Band, and Blue Note Bounce, score 3/5, 2/5, and 4/5 respectively; the two MusicGen pieces sit at the very bottom of the ratio range (1.071, 1.108) and score 2/5 and 3/5. The measured ratio orders none of this.

Improvisational coherence. The local-strong / global-weak split runs through this axis. Blue Dorian Drift is "macro-incoherent over four minutes but micro-level sounds good"; Whispers of the Dorian Sky (1/5) devolves into runs no longer fitting the chords by 0:55; and Chasing the High Flight (2/5) has a solo the rater calls "sporadic and random." Yet local

coherence can reach genuine solo construction: Sprint of the Brass scores 4/5 here inside a mid-scoring piece, on a xylophone solo at 1:53 with chromatic slides and motivic development, and Lush Chord Dreams' solo is judged "comparable to a real jazz pianist."

Idiomatic vocabulary. Scores range 2–5. Sprint of the Brass earns the study's only 5/5 — fully idiomatic phrasing and articulation — inside a 20/30 piece, showing vocabulary can be mastered locally while the piece fails structurally. Whispers of the Dorian Sky (2/5) is the mirror case: real chromatic and bebop shapes against a slow modal context where they do not fit. Latin Bop (3/5) is a style-specific variant — the surface of Latin jazz is present, but the clave "is not properly developed," leaving the idiom unconvincing at its core.

Ensemble interaction. The strongest axis (mean 3.7). Four pieces score 5/5: Rail Yard Bop; Corner Pocket Riff, whose big-band call-and-response evokes Count Basie; Blue and Purple, where an audible piano response at 2:04 is "the defining feature of this piece" despite a 20/30 total; and Lush Chord Dreams. Even weaker pieces can carry this axis: The Flaming Band scores 4/5 here — call-and-response is its single strongest feature — inside a 17/30 total. At the other pole, Chasing the High Flight (2/5) has percussion static throughout with zero call-and-response, and Whispers of the Dorian Sky (2/5) has parts that cease to follow one another. Interaction, like vocabulary, is separable from overall quality.

Formal structure. The weakest axis (mean 2.3) and the floor on quality. Corner Pocket Riff is the first piece to fully execute a form (5/5, a correctly repeated big-band A section), showing formal control is reachable rather than categorically absent, with Rail Yard Bop (4/5) the only other piece above 3. Nine of fourteen rated pieces score 2 or below, failing through absent B sections (Blue Dorian Drift), abrupt endings (Sprint of the Brass, Lush Chord Dreams), static repetition (Latin Bop), or near-total formlessness (Whispers of the Dorian Sky and Chasing the High Flight, both 1/5, and both MusicGen pieces, also 1/5).

4.4 Correspondence Between Automated Metrics and Expert Judgment

With all fourteen rated pieces, the two divergences can be characterised more precisely — and in one respect corrected.

The swing ratio does not track perceived swing, and fails worst at the extremes. The preliminary three-piece sample showed measured ratio ordering pieces in exact reverse of their swing scores. Across all fourteen the relationship is no longer a clean inversion, but the stronger claim survives: the measured ratio bears no reliable positive relationship to perceived swing and misleads most at the extremes of its range. Four observations establish this.

1. The two pieces scored 5/5 for swing sit near the bottom of the measured range — Corner Pocket Riff at 1.096 ("essentially straight," 404 onset pairs) heard as "essentially perfect" big-band swing, and Rail Yard Bop at 1.159 ("weak/light," 681 pairs) heard as "pretty much perfect" bebop — confirming that onset detection underestimates swing carried by the rhythm section collectively rather than by exposed melodic onsets.

2. The top of the range does not order with perceived swing either. The three highest ratios belong to Blue Dorian Drift (1.710, scored 3/5), The Flaming Band (1.786, scored 2/5), and Blue Note Bounce (2.014, scored 4/5). Blue Dorian Drift's high mean and variance track its abrupt 1:34 break rather than depth; The Flaming Band's ratio rests on just 36 valid onset pairs — against 238 to 785 for the reliable measurements — and the rater hears its swing collapse outright at 2:45, so the reading is mostly noise; and while Blue Note Bounce is the one high-ratio piece whose swing the rater credits (4/5), even there the number misses what the ear catches, since its swing is "mechanically perfect," with fills identical throughout and "too consistent to feel human." A high ratio can therefore mean genuine hard swing, an artefact of too few onsets, or a robotic regularity that reads as non-human — and the ratio alone cannot tell them apart.

3. The metric is informative only mid-range, where Moonlit Voicings (1.513, scored 4) and Lush Chord Dreams (1.487, scored 4) match the ear.

4. A low ratio is not always wrong, and near-identical low ratios can mean opposite things. Sprint of the Brass (1.105) genuinely is "too relaxed for the style" and scores 3/5, and the two MusicGen pieces sharpen the point: musicgen_blue (1.071), whose low ratio and low timing variance (sd 0.371) together capture genuinely metronomic, straight playing the rater confirms has "no convincing swing feel" (2/5), and musicgen_green (1.108), an almost identical ratio, which scores 3/5 because a Latin-inflected rhythm section carries a feel the onset ratio cannot see. Across the five pieces clustered near 1.05–1.16 — Latin Bop, musicgen_blue, Sprint of the

Brass, musicgen_green, Corner Pocket Riff, Rail Yard Bop — human swing scores run the full span from 2 to 5, so the same low reading can mean metronomic absence, an appropriate straight feel, ensemble swing the metric misses, or a measurement artefact. Latin Bop adds a style-level caution: its 1.047 — the most reliable measurement in the set (785 pairs, sd 0.268) — is correct, since straight eighths suit Latin jazz, yet the clave the style actually demands is invisible to onset-ratio measurement.

ii–V–I detection fails on polyphonic audio, predominantly but not only by under-detection. The detector reports two ii–V–I progressions across fourteen pieces, against a rated set in which the rater hears functional motion in at least six where the detector found none (Rail Yard Bop, "many ii-V-Is"; Blue Dorian Drift; Moonlit Voicings, 3–2–5–1 from 0:01, the detector's count of one an undercount; Lush Chord Dreams; Corner Pocket Riff, present though generic; and Blue Note Bounce, whose Cm7→F7 the rater hears as a ii–V in B ♭). The dominant error is under-detection, consistent with harmonic smearing defeating template matching on mixed audio. But the expanded sample corrects the preliminary claim that the tool "never over-detects": one of its two hits falls in Chasing the High Flight, where the rater hears "essentially no chord progression" — correct notes never assembled into motion. In some cases the two agree: on The Flaming Band and both MusicGen pieces, detector and ear alike find no functional ii–V–I. But the MusicGen pieces expose a different instrument's failure entirely — musicgen_blue pairs near-top harmonic metrics (97% complexity, 0.983 pitch-class similarity) with a 1/5 harmonic-authenticity score, the widest gap in the study between what the distributional metrics report and what the ear finds, which Section 5.3 takes up. The ii–V–I template can thus miss cadences the ear hears, flag one it does not, and occasionally get it right; and the distributional harmonic metrics can be maximal while functional harmony is absent — because neither instrument has access to whether harmony resolves, which is a relationship over time. Section 5 develops both divergences.

4.5 Detectability Verdicts

For each rated piece the rater also recorded whether, and by whom, the output would be identified as machine-generated. At the top of the range, Rail Yard Bop (26/30) would not be identified by a non-specialist, and a jazz musician would more likely call it "bad" than "AI".

Moonlit Voicings (23/30) "would fool most listeners" but remains detectable to a trained ear; and Blue Note Bounce (21/30) "would fool casual listeners," its swing macro-coherent in a way rare in this dataset, though a jazz musician catches it through fills identical throughout — a regularity too perfect to be human. Lower still, The Flaming Band (17/30) is detectable by casual listeners and, with its swing collapse at 2:45, definitive for jazz musicians. At the bottom sit four pieces identifiable by essentially everyone: Chasing the High Flight (13/30) and musicgen_green (13/30) are detectable as AI by most listeners, and Whispers of the Dorian Sky (10/30) and musicgen_blue (10/30) are identifiable by trained and untrained listeners alike, breaking down within the first phrase — musicgen_blue in particular the rater calls "sheer chaos … detectable as AI by anyone." These verdicts are interpreted in Section 5.4.

5 Discussion

5.1 What the Models Have Learned

Locally, the outputs are indeed harmonically fluent. Across all fourteen analysed pieces, jazz pitch-class similarity never falls below 0.91 and seventh-or-richer chords range from 63% to 98%, at or above 90% in eight. This is consistent with the rater's judgement: they assigned harmonic authenticity of 4/5 to six rated pieces and heard genuine functional ii–V–I motion in several — Gm7 → C7 → Fm7 and Fm7 → B ♭ 7 → E ♭ in Blue Dorian Drift, 3–2–5–1 from the downbeat in Moonlit Voicings, ii–V–I as a defining feature of Rail Yard Bop, and a near-complete C-minor functional palette in Lush Chord Dreams. The models have learned not just jazz-coloured vocabulary but, at least intermittently, jazz harmonic function. This is intermittent and not universal across systems: the MusicGen pieces reproduce the vocabulary — musicgen_blue posts near-top harmonic metrics — while producing no functional motion the ear can follow (1/5), a gap Section 5.2 develops.

Local micro-competence extends past harmony. Rail Yard Bop (26/30) earns 5/5 for both swing and ensemble interaction; Corner Pocket Riff (25/30) earns 5/5 for swing, ensemble, and — uniquely — formal structure, executing a correctly repeated big-band A section. Where the

models succeed they succeed at the phrase and texture level: harmony that resolves, timbre that reads as jazz, comping and walking bass in their proper roles, solos that are locally shapely. The statistical texture of jazz over short spans is well within reach.

5.2 What the Models Have Not Learned

The failures are structural and temporal, and consistent. Formal structure is the weakest axis (mean 2.3 across rated pieces) and the floor on overall quality: no piece scoring 2 or below on form exceeds 23/30, and the two pieces above 3 on form — Corner Pocket Riff (5/5) and Rail Yard Bop (4/5) — are the two highest-scoring overall. Corner Pocket Riff shows formal control is within reach rather than categorically absent; but nine of fourteen rated pieces still score 2 or below, failing through absent B sections (Blue Dorian Drift), abrupt endings (Sprint of the Brass, Lush Chord Dreams), or near-total formlessness (Whispers of the Dorian Sky, Chasing the High Flight, and both MusicGen pieces, all 1/5).

Underlying this is a recurring local/global split. Blue Dorian Drift is "macro-incoherent over four minutes but micro-level sounds good," with an unmotivated modal-to-swing shift at 1:34 and back at 3:30. Whispers of the Dorian Sky stages the same failure in miniature: a coherent opening that collapses into swing breakdown at 0:32, random harmony at 0:47, and runs no longer fitting the changes by 0:55. Idiomatic knowledge is misapplied at the structural level — Whispers' fast runs contain real chromatic and bebop shapes, but against a slow modal context they are correct vocabulary in the wrong place. The models appear to lack a representation of where a piece is in its own form and where it is going.

This is what one would predict from how the systems are built. Autoregressive generation over audio or symbolic tokens optimises the local transition — the next token given recent context — which suits the short-range regularities that make harmony, timbre, and phrasing sound right. It offers no explicit mechanism for the hierarchical, long-range dependencies that define form: the return of a head, the arc of a solo, commitment to a key or feel across minutes. High local competence with weak global structure is therefore not incidental but consistent with a genuine representational gap. Models use the local statistics of jazz without the hierarchy that organises them.

5.3 The Measurement Problem

A second, methodological finding runs alongside the first: the standard automated measures diverge from expert perception in systematic, interpretable ways on dense polyphony. This is not noise to average away but a result about the instruments.

The methodological warning is concrete, and musicgen_blue is its sharpest illustration: it scores at or near the top of every distributional harmonic metric in the study — 97% jazz complexity, 0.983 pitch-class similarity — while a trained ear rates its harmony the worst in the dataset (1/5). Off-the-shelf audio metrics — classical MIR tools like these, or distributional scores such as Fréchet Audio Distance — should be treated as diagnostics validated against expert judgment, not ground truth, because they miss exactly the competence distinctions a trained listener makes. Directions for future work include source separation to recover cleaner per-instrument signals, spectrogram-based or learned chord models in place of fixed templates, swing characterised by its full time-series and variance rather than a single mean, and hybrid evaluation pairing automated measurement with expert rubric assessment so each checks the other.

5.4 The Competence Ceiling

The most telling observation is how the best output fails. A jazz musician hears Rail Yard Bop not as "AI-generated" but as "bad" — generic, competent jazz below professional quality. The models have crossed the threshold of plausibility: their better outputs are no longer obviously synthetic; it is believable that they are human-created. They have not crossed the threshold of competence: formal commitment, long-range coherence, genuine interaction, and swing that lives in ensemble interplay remain absent or unreliable — and these are precisely the qualities audible chiefly to trained listeners. A lay-listener study of these same pieces would substantially overstate their competence, consistent with prior findings that experts distinguish AI jazz far more reliably than non-experts.

5.5 Implications for AI Music Evaluation

There are three implications from this study.

1. lay-listener ratings and distributional metrics are insufficient alone for claims about genre competence: both register plausibility and miss the structural and idiomatic failures that separate plausible from competent jazz.

2. Expert rubric-based evaluation is necessary to surface those failures, but must be disciplined — anchored to explicit criteria, justified against timestamps, and paired with a reliability check (ideally inter-rater at scale) so expert judgment is reproducible rather than idiosyncratic.

3. quantitative metrics remain valuable but belong as diagnostics validated against expert judgment, not substitutes; where the two disagree, as with swing and ii–V–I here, the disagreement is itself informative.

Overall, the results argue for making formal structure and long-range coherence explicit, first-class evaluation criteria, since these are both the most consistent weakness of current models and the dimension least visible to the lay-listener and distributional methods that dominate practice.

5.6 Ethical and Intellectual Property Considerations

Evaluating commercial AI music generators is not a neutral act, and three considerations deserve acknowledgement. First, provenance: the systems studied here are trained on large music corpora whose licensing is in several cases opaque or contested, and benchmarking their output should not be read as an endorsement of how that training data was obtained. Second, validation: by measuring quality at all, a study of this kind risks lending legitimacy to the systems it examines. We therefore frame our contribution as diagnostic rather than promotional — the central finding is that these systems fall short of competent jazz on the dimensions a trained musician values, which makes any claim of parity with human musicians premature on this evidence. Third, artist impact: outputs that pass as amateur human jazz raise real questions of displacement and attribution for working musicians, particularly as such audio becomes cheap to produce at scale.

For the study's own materials, the generated audio is released only to make the evaluation reproducible; the models' output licences vary — some are non-commercial — and we make no commercial use of any generated piece. We take no position here on the larger legal questions of training-data copyright, which remain unsettled, but note that an evaluation framework such as ours can serve the ethical debate as well as the technical one: it gives musicians and the public a

structured, transparent account of what these systems can and cannot yet do, rather than leaving that judgment to marketing or to lay first impressions.

6. Conclusion

This paper tested whether current AI music systems produce jazz that is genuinely competent or only superficially plausible, using a trained jazz musician rather than lay listeners or metrics alone. In our current sample a consistent picture emerges.

The systems have learned the local surface of jazz well and its long-range structure poorly. Harmonic content is uniformly rich and jazz-aligned, and the best outputs convincingly reproduce functional harmony, idiomatic swing, and reactive ensemble roles at the phrase and texture level. What the models do not reliably produce is form: formal structure was the weakest axis and the closest correlate of overall quality, and several pieces that open coherently degrade over their duration. This local-strong, global-weak pattern is what one expects from systems that optimize short-range token transitions without an explicit representation of hierarchical form, and it locates the current limit not in surface realism but in structure.

The diagnostic value of the failures is the point. Which axes fail, and how, shows what the systems do not yet understand: large-scale form, long-range coherence, and the ensemble interplay that produces swing. The best output fails revealingly: a musician hears it not as machine-generated but as merely mediocre, so current systems have crossed the threshold of plausibility for general listeners while staying below the threshold of competence for expert ones. That gap is invisible to the lay-listener and distributional methods that dominate practice — the central reason expert-led assessment is needed.

A second contribution is a caution about measurement. The standard tools — onset-based swing-ratio estimation and template-based ii–V–I detection — diverged from expert judgment systematically on dense polyphonic audio, to the point where measured swing bore no reliable relationship to perceived swing and the ii–V–I detector both missed real cadences and flagged an absent one. These divergences are findings in their own right: off-the-shelf audio metrics should be treated as diagnostics validated against expert judgment, not ground truth, and their failures point toward source separation, learned harmonic models, variance-aware swing measures, and hybrid human–automated evaluation.

As AI music becomes something millions encounter and assume to be good, evaluation that can hear the difference between plausible and competent is not a refinement but a requirement.

Use of AI Tools

Generative AI assistance (Anthropic's Claude Opus 5) was used for drafting and editing support during manuscript preparation, and for exploratory commentary within the per-piece analysis reports. All research design, prompt construction, audio generation, expert rubric ratings, musical judgments, quantitative analysis, and conclusions are the author's own. All AI-assisted text was reviewed, revised, and verified by the author, who takes full responsibility for the content of the manuscript.

Reproducibility

All prompts, generated audio, analysis code, per-piece reports, and rubric assessments supporting this study are openly available at https://doi.org/10.5281/zenodo.21977580. The repository contains the fixed prompt texts (PROMPTS.md), the full anchored rubric (methodology.md), the analysis pipeline, and the per-piece reports and figures underlying Tables 1 and 2. The generated audio is released solely to make the evaluation reproducible; see Section 5.6 regarding output licences.

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Competing Interests

The authors have no competing interests to declare.

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This work relies on the open-source scientific Python ecosystem. The analysis pipeline was built with librosa (McFee et al., 2015) for onset detection and constant-Q chroma extraction, together with NumPy, SciPy, and Matplotlib.

Author's Contributions.

The author designed the study, wrote the analysis pipeline, generated all evaluated material, conducted the expert rubric assessment, and wrote the manuscript.

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